Design of Experiments

Treatments, units and factors, experimental error, Fisher's three principles and ways to improve precision.

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1 Why design experiments?

Design of Experiments (DoE): planning experiments so treatment effects are estimated **accurately, efficiently and without bias**.

Objectives: validity · reliability · replicability · optimality.

A good design gives unbiased estimates with their uncertainty, detects important differences, includes the analysis plan, is easy to interpret, has wide validity, uses minimal resources and is as simple as possible.

2 Key terms

Treatment	Any object of comparison
Control	Standard treatment for comparison
Experimental unit	Smallest unit to which a treatment is **independently** applied
Observational unit	Unit on which the response is measured
Response	Measured outcome
Factor / levels	Variable studied / its settings

3 Experimental error

Variation among experimental units **receiving the same treatment**, caused by **extraneous factors**.

Every design must provide a way to **estimate** it.

4 Fisher's three basic principles

Principle	What it means	Why
Randomization	Treatments assigned to units purely by chance	Protects against allocation bias; balances unknown sources of variation on average
Replication	Same treatment on more than one unit	**Estimates experimental error**; improves precision and reliability
Local control (blocking)	Group similar units into blocks before randomising	**Reduces** experimental error from known variation

In a **Randomized Block Design**, all treatments are randomly assigned within each block of similar units.

5 Improving precision

- ✓ **Border effect:** neighbouring plots affect border plants, so record only the inner **net plot**.
- ✓ **Proper plot technique:** keep everything except the treatment uniform.
- ✓ **Appropriate analysis:** e.g. ANCOVA adjusts for covariates.
- ✓ **Planning and recording:** block for time or batches; keep original records.
- ✓ **Matching the design** to the experimental units.

6 History

- **R.A. Fisher** founded the statistical principles of DoE at **Rothamsted Experimental Station** in the 1920s–30s.
- He introduced **ANOVA** to partition variation into factors and chance.
- With **Frank Yates**, he built methods still central to agricultural research.
- Fisher's **tea-tasting** experiment shows why randomization matters.

7 Example: pot experiment with manures

Treatments	Poultry manure, cow dung, coir-pith compost
Experimental unit	One pot receiving one manure
Replication	Several pots per manure
Response	Yield per plant
Error	Yield differences among pots given the same manure

! Common mistakes

- × **Pseudo-replication:** counting several plants in one pot as replicates.
- × Randomizing the field but always applying treatments in the same order.
- × Recording border plants.

? Fill in the blanks

1. Applying each fertiliser dose to four plots uses the principle of _____. **[replication]**
2. Assigning treatments to plots by drawing lots is _____. **[randomization]**
3. Grouping plots of similar fertility into blocks applies _____. **[local control]**
4. Yield differences between plots that received the same treatment measure experimental _____. **[error]**
5. Counting three plants in one pot as three replicates is the mistake called _____. **[pseudo-replication]**
6. Border rows are left out and yield is recorded from the _____ plot. **[net]**

Source: *Textbook of Agricultural Statistics*, Ch. 16 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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